

✓ Scatter Plot (plt.scatter)

Purpose

Scatter plots are used to show:

- Relationship between two numerical variables
 - Correlation (positive / negative)
 - Clusters and patterns
 - Outliers
-

When to Use

Use scatter plots when:

- You want to compare two variables
- Data is not continuous like time series
- You want to check dependency between variables

Example:

Study Hours vs Marks

Height vs Weight

Syntax

```
plt.scatter(x, y)
```

Important Parameters

```
plt.scatter(x, y, color=None, marker='o', s=20)
```

color

Defines color of points

Examples:

- 'red'
 - 'blue'
 - 'green'
-

marker

Shape of points

Examples:

- 'o' → circle
 - 'x' → cross
 - '^' → triangle
-

s

Size of points

Default = 20

Default Behavior

- Color → blue
 - Marker → circle
 - Size → small
-

Key Understanding

Unlike line plots:

- Points are NOT connected
- Each point is independent
- Focus is on relationship, not trend

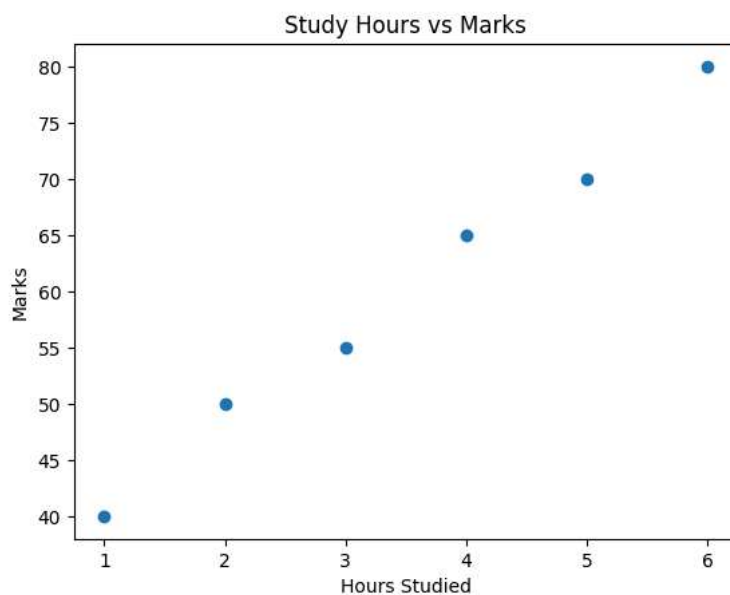
```
import matplotlib.pyplot as plt

# Sample data (study hours vs marks)
hours = [1, 2, 3, 4, 5, 6]
marks = [40, 50, 55, 65, 70, 80]

# Scatter plot
plt.scatter(hours, marks)

# Labels and title
plt.title("Study Hours vs Marks")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")

# Display
plt.show()
```



✓ Bar Chart (plt.bar)

Purpose

Bar charts are used to show:

- Comparison between categories
- Differences in values
- Ranking of items

When to Use

Use bar charts when:

- Data is categorical
- You want to compare multiple groups
- Order or ranking is important

Example:

Sales by product
Students in different departments
Revenue by region

Syntax

`plt.bar(x, height)`

Important Parameters

`plt.bar(x, height, color=None, width=0.8)`

x

Categories (labels on X-axis)

Example:

`['A', 'B', 'C']`

height

Values corresponding to each category

Example:

`[100, 200, 150]`

color

Defines color of bars

Examples:

- 'blue'
 - 'green'
 - 'orange'
-

width

Width of bar

Default = 0.8

Default Behavior

- Color → blue
 - Bars are evenly spaced
 - Width = 0.8
-

Key Understanding

- Each bar represents a category
- Height of bar = value
- Best for comparison, not trends

```
import matplotlib.pyplot as plt

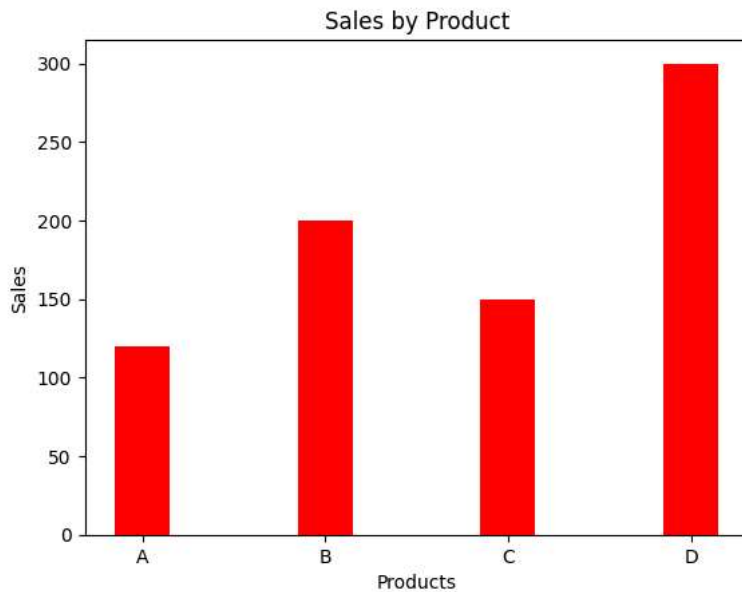
# Sample data (product vs sales)
products = ['A', 'B', 'C', 'D']
sales = [120, 200, 150, 300]

# Bar chart
plt.bar(products, sales, color = 'Red', width = 0.3)

# Labels and title
plt.title("Sales by Product")
plt.xlabel("Products")
```

```
plt.ylabel("Sales")
```

```
# Display  
plt.show()
```



✓ Histogram (plt.hist)

Purpose

Histograms are used to show:

- Distribution of numerical data
- Frequency of values
- Spread of data

When to Use

Use histograms when:

- Data is continuous
- You want to understand distribution
- You want to detect skewness or outliers

Example:

Exam scores

Age distribution

Income distribution

Syntax

```
plt.hist(data)
```

Important Parameters

```
plt.hist(data, bins=10, color=None, edgecolor=None)
```

data

Numerical dataset

bins

Number of intervals

Default = 10

Higher bins → more detailed distribution

Lower bins → more generalized

color

Color of bars

edgecolor

Border color of bars

Default Behavior

- Bins = 10
- Color = blue
- Shows frequency count

Key Understanding

- Groups data into ranges (bins)
- Shows how many values fall into each range
- Helps understand spread and shape of data

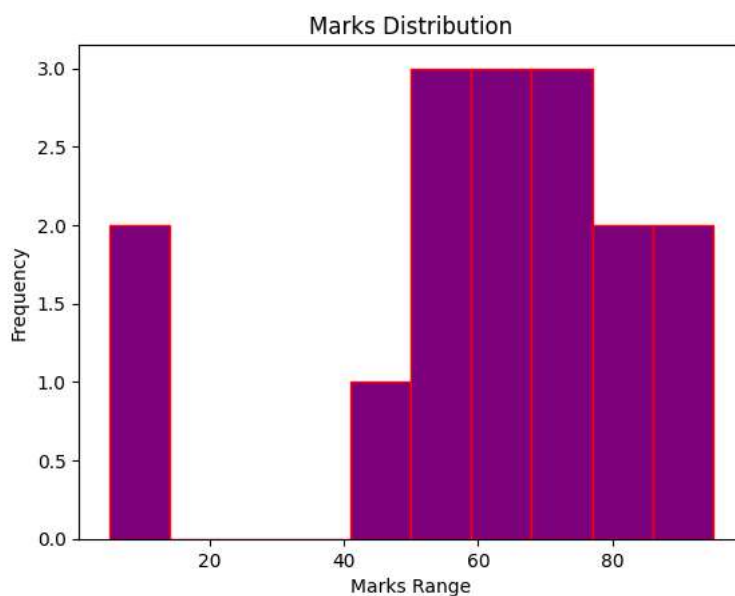
```
import matplotlib.pyplot as plt

# Sample data (marks distribution)
marks = [5,6,45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 50, 60, 70]

# Histogram
plt.hist(marks, bins=10, color = "purple", edgecolor = 'Red')

# Labels and title
plt.title("Marks Distribution")
plt.xlabel("Marks Range")
plt.ylabel("Frequency")

# Display
plt.show()
```



✓ Figure and Axes Concept

Understanding this is very important.

Figure

The entire canvas where everything is drawn.

Think of it as:

- The full page
-

Axes

The actual plotting area inside the figure.

Think of it as:

- The graph area
-

Axis

Refers to:

- X-axis (horizontal)
 - Y-axis (vertical)
-

Structure

Figure → contains Axes

Axes → contains Axis

Why This Matters

- Helps in creating multiple plots
 - Needed for subplots
 - Gives better control over plots
-

Basic Syntax

```
fig, ax = plt.subplots()
```

```
ax.plot(x, y)
```

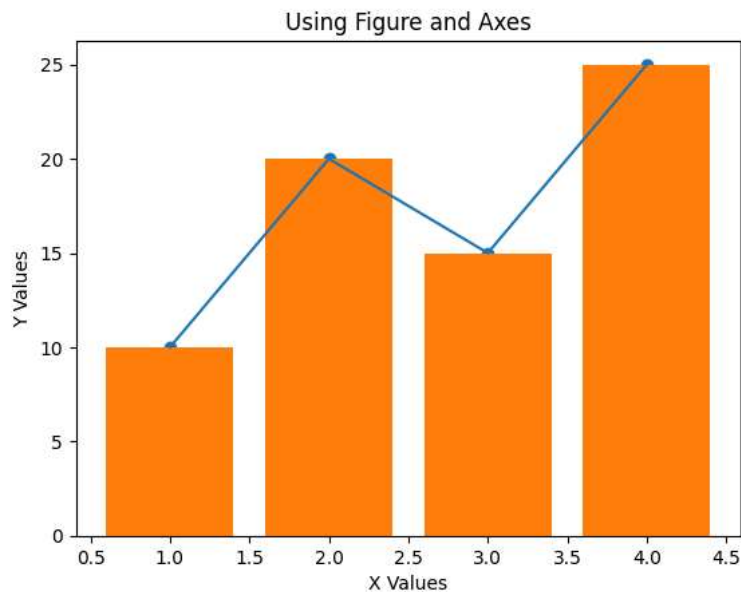
```
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3, 4]
y = [10, 20, 15, 25]

# Create figure and axes
fig, ax = plt.subplots()
# fig.plot(x,y)
# Plot using axes object
ax.plot(x, y)
ax.scatter(x,y)
ax.bar(x,y)

# Labels and title
ax.set_title("Using Figure and Axes")
ax.set_xlabel("X Values")
ax.set_ylabel("Y Values")

# Display
plt.show()
```

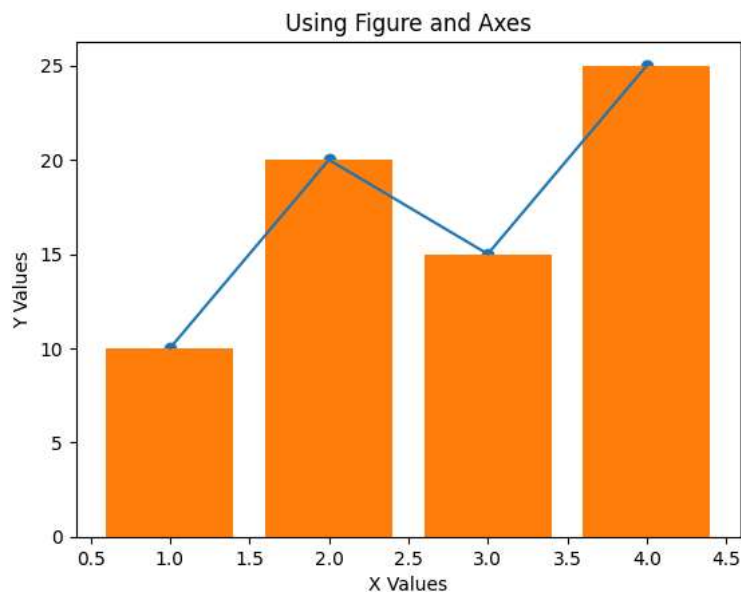


```
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3, 4]
y = [10, 20, 15, 25]

plt.plot(x, y)
plt.scatter(x, y)
plt.bar(x, y)
# Labels and title
plt.title("Using Figure and Axes")
plt.xlabel("X Values")
plt.ylabel("Y Values")

# Display
plt.show()
```



Customizing Charts

Customization improves readability and presentation.

Common Customizations

- Title
 - X-axis label
 - Y-axis label
 - Legend
 - Colors
-

Syntax Overview

```
plt.title("Title")
plt.xlabel("X Label")
plt.ylabel("Y Label")
plt.legend()
```

Legend

Used when multiple plots are present.

Requires label parameter in plot.

Example:

```
plt.plot(x, y, label="Line 1")
plt.legend()
```

Key Understanding

- Always label your charts
- Use meaningful titles
- Legends help in multi-line plots

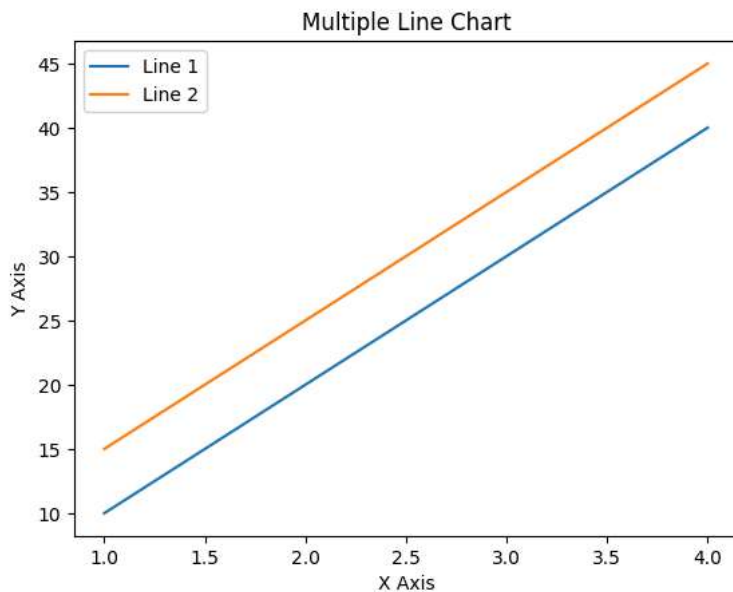
```
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3, 4]
y1 = [10, 20, 30, 40]
y2 = [15, 25, 35, 45]

# Plot multiple lines
plt.plot(x, y1, label="Line 1")
plt.plot(x, y2, label="Line 2")

# Customization
plt.title("Multiple Line Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.legend()

# Display
plt.show()
```



✓ Saving Charts

Sometimes we need to save plots instead of just displaying them.

Syntax

```
plt.savefig("filename.png")
```

Important Points

- File is saved in current working directory
- Supported formats:
 - .png
 - .jpg
 - .pdf

Best Practice

Always call `savefig()` BEFORE `show()`

Example Use Cases

- Reports
- Presentations
- Dashboards

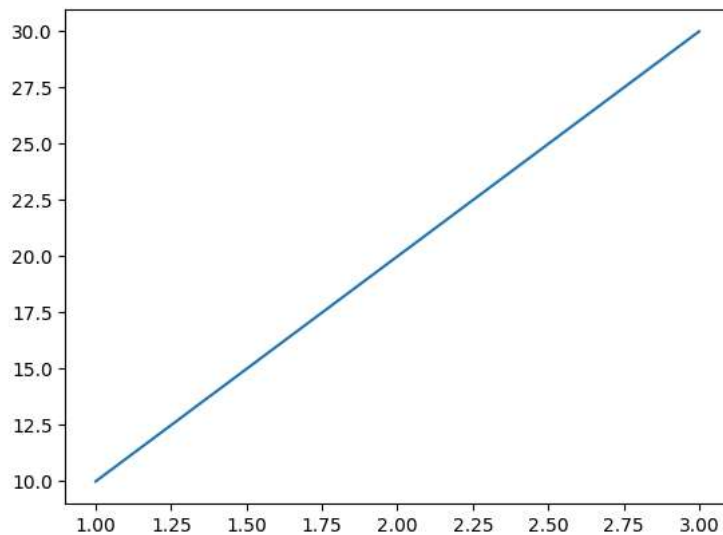
```
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3]
y = [10, 20, 30]

# Plot
plt.plot(x, y)

# Save figure
plt.savefig("line_plot.png")

# Display
plt.show()
```



Question 1

Create a scatter plot using:

```
hours = [1, 2, 3, 4, 5, 6]
```

```
marks = [40, 50, 55, 65, 70, 80]
```

Tasks:

- Plot the data
- Add title "Study Hours vs Marks"
- Label both axes properly

Question 2

Modify the scatter plot:

- Change color to red
- Change marker to 'x'
- Increase size of points

Question 3

Create a scatter plot using:

```
age = [20, 22, 24, 26, 28]
```

```
income = [20000, 25000, 30000, 35000, 40000]
```

Tasks:

- Plot the relationship
- Add labels and title

Question 4

Create a bar chart using:

```
products = ['A', 'B', 'C', 'D']
```

```
sales = [100, 150, 80, 200]
```

Tasks:

- Plot the chart
- Add title and axis labels

Question 5

Modify the bar chart:

- Change color to green
 - Adjust width of bars
 - Try using different category names
-

Question 6

Create a bar chart for:

```
departments = ['CS', 'IT', 'ECE', 'ME']
```

```
students = [120, 100, 80, 60]
```

Tasks:

- Plot the data
 - Add meaningful labels
 - Customize color
-

Question 7

Create a histogram using:

```
marks = [45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 50, 60, 70]
```

Tasks:

- Plot the histogram
 - Use default bins
 - Add title and labels
-

Question 8

Modify the histogram:

- Change number of bins
 - Change color of bars
 - Add edgecolor
-

Question 9

Create a histogram for:

```
ages = [18, 19, 20, 21, 22, 23, 24, 20, 21, 22, 23, 24, 25]
```

Tasks:

- Plot distribution
 - Experiment with bins
 - Add labels and title
-

Question 10

Combined Practice Task:

Using any dataset of your choice:

- Create one scatter plot
- Create one bar chart
- Create one histogram

Ensure:

- Proper titles
 - Proper labels
 - At least one customization in each plot
-

Important Instruction

- Use `plt.show()` after every plot
- Keep code clean and readable
- Focus on understanding, not just execution # Practice Questions — Matplotlib (Scatter Plot, Bar Chart, Histogram)

These questions are designed to strengthen your understanding of:

- Scatter Plot
 - Bar Chart
 - Histogram
-

Question 1

Create a scatter plot using:

hours = [1, 2, 3, 4, 5, 6]

marks = [40, 50, 55, 65, 70, 80]

Tasks:

- Plot the data
 - Add title "Study Hours vs Marks"
 - Label both axes properly
-

Question 2

Modify the scatter plot:

- Change color to red
 - Change marker to 'x'
 - Increase size of points
-

Question 3

Create a scatter plot using:

age = [20, 22, 24, 26, 28]

income = [20000, 25000, 30000, 35000, 40000]

Tasks:

- Plot the relationship
 - Add labels and title
-

Question 4

Create a bar chart using:

products = ['A', 'B', 'C', 'D']

sales = [100, 150, 80, 200]

Tasks:

- Plot the chart
 - Add title and axis labels
-

Question 5

Modify the bar chart:

- Change color to green
 - Adjust width of bars
 - Try using different category names
-

Question 6

Create a bar chart for:

```
departments = ['CS', 'IT', 'ECE', 'ME']  
students = [120, 100, 80, 60]
```

Tasks:

- Plot the data
 - Add meaningful labels
 - Customize color
-

Question 7

Create a histogram using:

```
marks = [45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 50, 60, 70]
```

Tasks:

- Plot the histogram
 - Use default bins
 - Add title and labels
-

Question 8

Modify the histogram:

- Change number of bins
 - Change color of bars
 - Add edgecolor
-

Question 9

Create a histogram for:

```
ages = [18, 19, 20, 21, 22, 23, 24, 20, 21, 22, 23, 24, 25]
```

Tasks:

- Plot distribution
 - Experiment with bins
 - Add labels and title
-

Question 10

Combined Practice Task:

Using any dataset of your choice:

- Create one scatter plot